



UNIVERSITY OF MORATUWA
Faculty of Information Technology

M.Sc / PG Diploma in Artificial Intelligence
2024 Batch 17 - Semester 4 Examination

IT 6903 – Scientific Communication

Time Allowed: 2 hours

, February 2026

INSTRUCTIONS TO CANDIDATES

1. This paper contains 4 questions on **3** Pages, including the cover page.
2. Total marks attainable for this examination is 100. The marks assigned for each question & sections thereof are included in square brackets.
3. This examination accounts for 70% of the module assessment.
4. This is an open-book examination.
5. Answer **ALL** questions.

Question 1

- (a) The ability to communicate has been one of the most influential factors in the evolution of humanity. In the modern era, human communication has extended beyond human–human interaction to include man–machine and machine–machine communication.
- (i) What is the distinctive feature of human communication compared to animal communication? Briefly explain how this feature has influenced the evolution of humanity.
 - (ii) With examples, identify technologies that enable the following types of communication: (p) man-man, (q) man-machine, (r) machine-machine.
 - (iii) “AI is approaching the technological singularity in language processing.” Do you agree with this statement? Justify your answer. [3x3 marks]
- (b) Which of the following is/are true? In each case, justify your answer.
- (i) Biological programming enables communication between plants and/or animals.
 - (ii) There is a synergy between recent developments in non-symbolic AI and symbolic AI in language processing.
 - (iii) Before the birth of AI, Alan Turing emphasized the role of communication in determining machine intelligence.
 - (iv) Syntax is the most critical concern when communicating algorithmically.
- [4x3 marks]
- (c) AI systems already exhibit many intelligent features. However, some cognitive features remain under development, and relationships among cognitive features have been identified by neuroscientists and educationists.
- (i) Identify one education-related cognitive feature that enhances other cognitive abilities.
 - (ii) Propose an AI-based solution to enhance this cognitive feature in humans and identify one possible negative implication.
- [4 marks]

Question 2

(a) Machines have surpassed human performance in certain aspects of reading comprehension. Reading remains a fundamental skill for acquiring knowledge.

(i) Identify two broad categories of information extracted from research papers. Describe additional information a novice researcher may gain by reading research papers.

(ii) A novice researcher uses an AI tool (e.g., Gemini) to summarize papers and directly writes a research proposal. Identify THREE immediate negative implications. Provide an analogy and explain how such tools should be used.

(iii) Discuss how the following techniques improve reading comprehension.

(i) Reading aloud

(ii) Reading highlighted text

(iii) Automating left-to-right eye movement

[3x3 marks]

(b) Identify which of the following is/are true. In each case, justify your answer.

(i) In a double-blind review process, reviewers independently send comments directly to authors.

(ii) Researcher credibility is assessed using multiple metrics derived from two main parameters.

(iii) Since knowledge is considered to be the power, a bigger portion of the finances involved in the publication process goes to authors who are the creators of knowledge.

(iv) Reading and writing are also identified as activities that naturally demand mindfulness.

[4x3 marks]

(c) Reading and writing are closely associated cognitive skills. Generally, all good writers are good readers.

- (i) What are the best practices in reading research papers for a research project?
- (ii) Identify the most influential best practice in the above that energizes at least three cognitive skills. Justify your answer, clearly indicating the three cognitive features.

[4 marks]

Question 3

(a) Any scientific document talks about two main topics. Under these two topics, researchers develop some content to elaborate on the topics. In a research process, a researcher produces several scientific documents, including a thesis.

- (i) Name FOUR major scientific documents in the research process. Outline the content of the 1st Scientific document produced by a researcher and indicate how the two topics are distributed in the scientific document.
- (ii) What is the most influential chapter for developing the entire thesis? Justify your answer.
- (iii) At which stage can a researcher conceptualize the entire project? Identify the content to be developed at this stage.

[3x3 marks]

(b) Identify which of the following is/are true. In each case, justify your answer.

- (i) When writing a thesis, the chapters should be carefully developed in the sequential order of chapters.
- (ii) In the Milestone Approach, the general term, Methodology, in a research process has been adopted to suit the steps in computing projects.
- (iii) For good reasons, both objectives and problem definition should come before the Literature Review in a thesis.
- (iv) Only a few Research Software offer facilities to develop all sections/chapters as per a research method.

[4x3marks]

- (c) The complex process of writing a thesis can be guided by a simple four-step framework: outline, draft, revise, and fine-tune. These steps are bidirectional and work sequentially.
- (i) Identify the people involved in the four stages of the above framework. Justify your answer by giving reasons.
 - (ii) Briefly explain how/why you would use AI tools in the writing process by ensuring your language skills are not negatively influenced.

[4 marks]

Question 4

- (a) A research process involves multiple evaluations during the project period. Generally, each evaluation has two components, although some institutions may adopt variations.
- (i) Name the major evaluation points in a research process. Briefly describe the purpose of scientific documents/presentations prepared for each evaluation.
 - (ii) What are the two components of evaluations at different stages of the research process? Identify different methods used by institutions to conduct the final thesis evaluation and comment on these approaches.
 - (iii) Identify situations in which a research student may face difficulties during evaluation. What possible judgments might examiners make in these situations?
- (b) Identify which of the following is/are true. Justify your answer in each case.
- (i) Both objectives and problem definition should appear after the Literature Review in any research presentation.
 - (ii) Slides for the Progress Review presentation can be developed simply by adding steps in the research process after the Approach.
 - (iii) During Q&A in a viva, it is inappropriate to use additional material such as notes beyond the prepared slides.
 - (iv) Viva-voce examinations assess more than the technical knowledge of the candidate.

[3x3 marks]

[4x3 marks]

- (c) Recent advances in AI have reduced the demand for certain human skills in many employment sectors, including customer service and IT-related roles.
- (i) Identify FIVE biological limitations of humans that have been superseded by the machine.
 - (ii) Discuss which human skill is most threatened by these developments in the future.

[4 marks]

END OF PAPER